

RESEARCH ARTICLE

Video-Centered Multimodal Learning for Non-Invasive Detection of Emotional Workload in Emotional Labor Settings

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ABSTRACT Workers in emotional labor environments often follow display rules that require them to express emotions as directed by their boss or the company, regardless of their true feelings. These workers suppress their true feelings, which leads to ongoing mental issues and an accumulation of emotional workload. For example, in a call center situation, if a customer is shouting and making accusatory comments at the worker, the worker should respond in a positive way to the customer, even if they have negative feelings. However, it still remains a difficult issue to estimate emotional workload in real-world settings using non-invasive methods. In this study, we propose a machine learning model that automatically detects workers' emotional workload in emotional labor situations using multimodal data, including video and physiological data. We used a dataset designed based on a call center scenario and estimated emotional workload using two labels: an objective label based on experimental conditions, and a subjective label based on workers' self-reports. As a result, we found that geometric features derived from facial expressions captured in videos significantly contributed to model performance and that applying Multiple Kernel Learning (MKL) improved classification accuracy, especially when detecting subjective emotional workload. This study demonstrates the potential of image-based and MKL-based models for scalable and non-invasive detection of emotional workload, suggesting that they can be effectively applied in real-world work settings.

INDEX TERMS Emotional workload, emotional labor, video-based detection, kernel learning.

I. INTRODUCTION

Emotion regulation refers to the cognitive strategies to reduce, maintain, or increase one's emotional impact [1], [2]. For example, in emotionally demanding environments such as customer service, individuals are often required to suppress their true negative feelings and exhibit positive emotional expressions instead. Such repeated suppression of negative emotions through sustained self-control can cause serious psychological and mental problems, such as emotional exhaustion and depression. Due to its significant impact on

mental health, emotion regulation has been a central topic in psychology, as well as in the field of Human-Computer Interaction (HCI) [3], [4], [5], [6]. Particularly, Park et al. [7] conceptualized emotion regulation in the workplace as emotional workload, and have proposed models for continuously monitoring it using physiological and audio data. Detecting emotion regulation is fundamentally different from emotion recognition in terms of the expressiveness of felt emotion. Conventional emotion recognition estimates current affective status from observable emotional displays that are aligned with their felt emotions [8], [9], [10]. In contrast, emotion regulation involves detecting internal emotional states that may not be outwardly expressed. Thus, more indirect cues

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(e.g., physiological change, facial expression, and vocal prosody) become crucial.

Similar to emotion recognition models, physiological data have been utilized to assess emotion regulation [7], [11], [12]. However, using physiological data has been limited by its invasiveness because users have to tightly wear or attach sensors to collect the data. Furthermore, physiological data can raise privacy concerns, as it may reveal information about a worker's health status. Such information could potentially be used to monitor productivity or influence hiring decisions. On the other hand, video data presents fewer concerns regarding invasiveness and privacy, as it enables non-contact data collection and does not directly capture health-related information.

Traditionally, when constructing emotion recognition datasets, participants are instructed to explicitly display specific emotions through their facial expressions. However, in a real-world environment that requires emotion suppression, people's facial expressions may not be explicitly revealed. For example, a grimace or a down-turned corner of the mouth may appear when a subject is experiencing negative emotions such as anger or sadness, whereas a polite smile may appear in situations where emotional labor is required to regulate one's emotions [13].

Motivated by this, we focus on the following two research questions, with the goal of developing models that automatically evaluate emotional workload in the emotional labor context using a non-intrusive approach:

- RQ1: How can we estimate the emotional workload from video data of the worker's face under emotionally suppressed conditions?
- RQ2: How can we improve the performance of detecting emotional suppression?

In RQ1, we aimed to verify whether it is possible to estimate a worker's emotional workload from video data capturing the worker's face in a situation where emotions are suppressed. Following this, RQ2 explores how to fuse video-related features and how to further combine them with other data such as audio and physiological data to better detect whether a worker is emotionally suppressed.

To answer RQ1, we used two types of features that can be extracted from video data: geometric features and behavioral features. Geometric features include the distance or angle between each point landmark on the face, which we extracted based on the previous study [14]. We also extracted the behavioral features based on research in stressful or emotionally suppressed situations [2], [15], [16], [17]. Furthermore, we performed feature importance analysis to identify which video-based features were most informative in detecting emotional workload.

To answer RQ2, we used Multiple Kernel Learning (MKL) [18], which can effectively integrate information of multiple features obtained from multimodal sensor data. MKL assigns modality- or feature-specific kernels and effectively combines them via kernel weights and kernel-specific

hyperparameters (e.g., gamma and C) to enable effective multimodal feature fusion. End-to-end deep learning-based multimodal fusion methods have also been explored [19], [20], but such approaches often require large-scale data and offer limited transparency in practice [21]. However, human-subject multimodal data are frequently limited in size due to collection cost and practical constraints, and different modalities may contribute differently to predictions because of their distinct characteristics. Therefore, we adopted MKL as a practical alternative for low-data settings and compared its performance with several widely used machine learning and deep learning models.

We utilized the Hide-and-seek dataset [7] which comprehensively collects physiological, audio, and video data from the customer actor and the subject, the worker on a call center scenario. A total of 31 subjects participated in the study. For labels, we used two main labels in the dataset: (1) Labels that are either suppressed or unsuppressed, depending on the experimental context (neutral, shouting, or swearing conditions) (i.e., Given-Emotional-Workload), and (2) Labels in which subjects self-reported their suppressed workload level (i.e., Perceived-Emotional-Workload), binarized based on whether their scores were above or below each subject's average.

To investigate the effectiveness of video-based features and the benefit of multimodal fusion, we proposed two modeling settings: (1) an all multimodal setting that integrates all available features from physiological, audio, and video data, and (2) a video-only setting that relies solely on video-derived features, providing a less intrusive, more practical solution for real-world deployment. For both modeling settings, we benchmarked seven widely used machine learning algorithms, including Support Vector Machine (SVM) and Random Forest (RF). In addition, we applied MKL to evaluate the potential of kernel-based multimodal fusion. We also evaluated modern image-based deep learning models (e.g., EfficientNet and CLIP) and an MLP model in the video-only setting. Additionally, we compared the video-only setting against models based on physiological features, which are widely used but require intrusive sensor attachment. Model performance was assessed using accuracy, balanced accuracy, F1-score, and Area Under the ROC Curve (AUC) score.

As a result, we found that MKL-based feature fusion exhibited the best performance in both modeling strategies (i.e., all multimodal and video-only settings). Specifically, in the multimodal setting, MKL achieved accuracy/balanced accuracy of 0.88/0.87 and 0.71/0.72 for the Given-Emotional-Workload and Perceived-Emotional-Workload labels, respectively. In the video-only setting, MKL achieved 0.67/0.59 and 0.63/0.63, respectively. Notably, for the Perceived-Emotional-Workload label, MKL improved both accuracy and balanced accuracy by 0.05, 0.04 points compared to SVM in the multimodal and video-only settings, demonstrating the effectiveness of kernel-based feature fusion. Moreover, the video-only MKL model outperformed

the physiology-based SVM model by 0.04 in both accuracy and balanced accuracy. Based on these results, we discussed the external validity and scalability of the proposed model in real-world work environments.

II. RELATED WORKS

A. MULTIMODAL APPROACHES FOR EMOTION RECOGNITION

In recent years, emotion recognition using multimodal data has become an important research area in HCI. For example, datasets such as K-EMOCON [22], WESAD [23], and the dataset collected by Soleymani et al. [10] include multiple modalities, such as video, physiological signals, and audio, which have been shown to improve the accuracy of emotion recognition systems.

Among the various modalities in multimodal emotion recognition, physiological signals are commonly used due to their ability to reflect internal bodily responses. For example, Schmidt et al. [23] collected physiological data using sensors attached to various body locations and presented a model to distinguish emotional states that achieved high classification accuracy. In addition, Yang et al. [24] proposed a deep multimodal emotion recognition system using behavioral and physiological signals from mobile and wearable devices. However, the physiological data collection approach is intrusive and impractical because it requires numerous sensors to be attached to the users, which causes inconvenience to users.

Non-intrusive methods using audio and video data have become increasingly used to solve the problem of intrusive physiological data collection in emotion recognition systems. Audio data has been widely used in many studies, as it captures key emotional cues, such as tone, pitch, and rhythm, without requiring physical contact [25]. For example, prior studies [8], [9] have shown that audio features provided a valuable insight into a person's emotional state. Similarly, video data is also commonly used in emotion recognition, as it enables the extraction of various features related to emotion recognition through image analysis and facial landmarks analysis [14], [26]. In particular, the video-based approach can capture not only macroscopic facial expressions but also fine facial expressions representing emotional states. Furthermore, video data has been shown to contribute to improving the accuracy of emotion recognition systems when combined with other modalities such as audio or physiological signals. For example, Song and Kim [27] and Brady et al. [28] achieved higher predictive performance by integrating video data and physiological signals.

Among these non-intrusive methods, video data stands out in emotion recognition due to its ability to extract a wide range of visual features through image analysis and landmark-based techniques. This approach is especially preferred in modern, user-friendly emotion recognition systems because this approach enables sophisticated emotion analysis without the inconvenience of attaching sensors.

B. FACIAL EXPRESSION ANALYSIS FOR EMOTION RECOGNITION

Many studies leverage facial expressions to recognize emotions. Specifically, faces are detected in the video data and key facial landmarks are extracted to derive facial features. Subsequently, these features are used to analyze facial expressions, train models, and evaluate models.

For face detection, methods such as Haar cascades provided by OpenCV or histogram of Oriented Gradients (HOG) features coupled with a linear Support Vector Machine (SVM), commonly used in the Dlib library, are employed [29], [30]. CNN-based models such as multitask cascades convolutional networks (MTCNN) are also utilized for face detection [31]. To extract key facial landmarks from the detected faces, the 68-point facial landmark detector provided by the Dlib library [32], or CNN-based networks (e.g., OpenFace [33], [34]) are often applied. These landmarks are then used to further extract detailed facial features. One prominent method for this is the Facial Action Coding System (FACS) [35], which codes facial muscle movements into action units (AUs). The extracted facial features are then analyzed using machine learning models such as SVM or RF [14]. Recent studies also leverage deep learning approaches for facial emotion recognition, particularly CNN-based models [29], [36], [37] that commonly adopt ResNet [38], [39], [40], [41] and EfficientNet [42], [43], [44] backbones. Image representation models such as CLIP-style visual encoders have also been applied to emotion recognition [45], [46], [47], [48].

In addition, some emotion recognition studies have directly collected geometric data. Bailenson et al. [49] identified the dominant features that significantly influence the performance of the emotion recognition model by gathering geometric data such as the average position of the left mouth corner, the ratio of mouth to eye aspect ratios, and mouth height. Geometric features such as distance and angle between facial landmarks, mainly around eyebrows, eyes and mouth, were also used as features [14]. In addition, Bhosale et al. [29] measured the distance between the lips and eyebrows to determine stress levels.

Similarly, emotional regulation studies have used features such as body movements, facial movements, mouth movements, yawning, smiling, crying, blinking, eye gaze, and facial touching to assess facial behavior and have shown significant results [2], [17]. However, these studies required subjects not to show any emotion and were not conducted in interpersonal situations. In the workplace, where emotional labor is required, workers must regulate their emotions in interpersonal situations as they must interact with customers. In addition, they are asked not only to hide their true emotions but also to express positive emotions regardless of their actual feelings [50]. This mismatch between the actual emotions of the worker and the emotions they are required to express can lead to mental problems if they perform emotional regulation for a long period of time [51]. Therefore, we conducted

experiments using these facial features to evaluate emotional regulation during interpersonal interactions.

C. ENHANCING EMOTION RECOGNITION WITH MULTIMODAL FUSION TECHNIQUES

To improve emotion recognition performance, a range of approaches have been proposed, particularly those based on multimodal fusion.

Beyond unimodal modeling, deep learning-based multimodal fusion approaches have integrated heterogeneous signals, such as facial expressions, physiological signals, audio prosody, and gaze, to improve emotion classification performance [8], [10], [24], [27], [52], [53], [54], [55]. Common fusion strategies include feature-level fusion (combining multimodal features before classification), decision-level fusion (combining predictions from separately trained unimodal models), and ensemble learning. For example, Seng et al. [52] showed that the feature-level fusion of visual and audio data significantly improved emotion recognition accuracy, and Soleymani et al. [10] demonstrated that the decision-level fusion of EEG responses and eye gaze data outperformed single-modality models. More recently, end-to-end multimodal deep-learning architectures have also been explored to learn fused representations from heterogeneous signals [19], [20]. While such models can be effective, they often require large-scale data and offer limited transparency [21].

Another effective approach for multimodal fusion in emotion recognition is Multiple Kernel Learning (MKL), which combines heterogeneous feature types more effectively by assigning modality- or feature-specific kernels and learning their optimal combination. Prior studies have demonstrated the effectiveness and robustness of MKL for emotion recognition in human-subject multimodal settings, even when data are limited [9], [56], [57], [58], [59]. Furthermore, MKL has been shown to yield significant gains by combining multiple kernels defined over subsets of facial features [60], [61] in unimodal settings. In addition, the MKL structure can offer a more transparent view of modality- or feature-level contributions and fusion mechanism.

III. DATA DESCRIPTION

This study is based on the Hide-and-seek dataset [7], which contains multimodal data from a simulated call center scenario involving customer actors and the worker participants. The customer actors consist of two females and two males, and there were 31 worker participants (24 females and 7 males), all of whom were currently employed as emotion workers at six companies operating call centers. The experimental design included three distinct call scenarios: neutral, shouting, and swearing. While the neutral condition served as a baseline, the shouting and swearing conditions were designed to induce emotional workload in the worker (see Figure 1).

During these interactions, researchers collected multiple types of data to comprehensively capture the emotional

dynamics involved. The dataset includes customer audio, worker audio, worker physiological signal, worker video, and self-reported scores that indicate the level of emotional suppression of the workers. The customer audio data was recorded as the situational cues intended to influence the workers' emotional responses during the calls. The worker audio data was recorded to assess vocal expressions that may indicate emotional inhibition, reflecting the emotionally regulated responses required in the situation. In addition, the worker physiological signal was collected using wearable devices, capturing signals such as electroencephalography (EEG), electrocardiogram (ECG), and electrodermal activity (EDA). These physiological signals provided insights into the workers' internal emotional states during the calls, revealing genuine emotional responses that may not be externally expressed. Moreover, front-facing videos of workers were recorded and can be analyzed for visual indicators of emotion, such as facial expressions and body language. In prior research using the Hide-and-seek dataset [7], only audio and physiological data were used to detect emotional workload. In contrast, to enhance the external validity and scalability of the model, we employed video data instead of invasive physiological signals that require sensor attachment to the user.

The data was labeled in two ways: (1) Given-Emotional-Workload: We categorized call situations into binary labels reflecting whether the scenario was neutral or stressful (shouting or swearing conditions) according to each call situation. (2) Perceived-Emotional-Workload: We categorized the scores into binary labels (high suppression vs. low suppression) based on the self-reported suppression scores of each participant, using their individual mean as the threshold.

IV. DATA ANALYSIS

A. FEATURE EXTRACTION

The audio and physiological data were pre-processed following the procedures described in [7], which used the same dataset. Specifically, audio and acceleration data were segmented into 10-second windows with a 1-second slide, and the remaining data (i.e., EDA, EEG, TEMP, and ECG) were segmented into 5-second windows. From each segment, a total of 228 features were extracted, including statistical features such as mean, standard deviation, minimum, and maximum, to capture key characteristics over time [62].

1) GEOMETRIC AND BEHAVIORAL FEATURES FROM VIDEO DATA

Previous studies on emotion recognition and emotion regulation have actively used geometric and behavioral features to extract visual information from video data [2], [14], [15], [16], [17], [29], [49]. Geometric features are derived by calculating the distances and angles between facial landmarks, and behavioral features capture facial movements or body movements. Based on these previous studies,

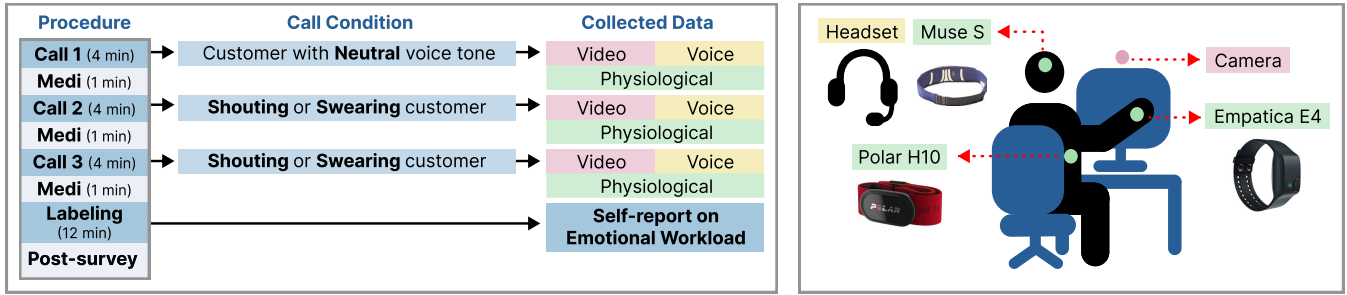


FIGURE 1. Data Collection Protocol. In a simulated call-center scenario, workers interacted with customers displaying various attitudes (neutral, swearing, shouting). During these interactions, customers' audio, workers' audio, video, and physiological data were collected. After the calling session, workers provided self-reported labels on how much effort they put into regulating their emotions.

TABLE 1. Geometric and Behavioral Features from Video Data. This table summarizes the video-based features extracted from facial landmarks using MediaPipe [63]. It presents 240 geometric features and 121 behavioral features, designed to capture fine-grained facial movements and expressions. These features include distances and angles between landmarks, as well as higher-level behavioral cues such as head movement and hand-to-face contact.

Data	Feature Group	# of Features	Description
Video	240 Geometric Features		
	<i>Distance and Angle between landmark numbers (Mediapipe). Mean, STD, Min, Max values for each feature</i>		
	[Distance] Eyebrows	56	70-63, 63-105, 105-66, 66-107, 336-296, 296-334, 334-293, 293-300, 70-33, 105-160, 107-133, 336-362, 334-387, 300-263
	[Distance] Eyes	48	33-160, 160-158, 158-133, 133-153, 153-144, 144-33, 362-385, 385-387, 387-263, 263-373, 373-380, 380-362
	[Distance] Mouth	48	61-39, 39-37, 37-0, 0-267, 267-269, 269-291, 291-321, 321-314, 314-17, 17-84, 84-91, 91-61
	[Distance] Nose	24	98-99, 99-194, 194-328, 328-327, 98-61, 327-291
	[Distance] Jaw	24	136-149, 149-148, 148-152, 152-377, 377-378, 378-365
	[Angle] Eyebrows	8	70-105-107, 336-334-300
	[Angle] Eyes	24	160-133-144, 33-160-133, 160-133-144, 387-263-373, 362-387-263, 387-362-373
	[Angle] Mouth	8	0-61-17, 0-291-17
	121 Behavioral Features		
	<i>Mean, STD, Min, Max values for each feature</i>		
	Distance-based behavioral features	32	Distance between eyebrows, Distance between eyes and eyebrows, Height of both eyes, Length of jaw, Width & Height of mouth, and Distance between mouth and eyes
	Body movement	8	Shoulder position changes and Shoulder angle changes
	Facial movement	4	Head tilt angle
	Facial touching	4	Distance between hand and face
	Eye movement	8	Changes in both eye positions
	Eye blinking	5	# of Eye blinking within window size
	Temporal features	60	Rate of changes in other behavioral features

we extracted 361 video features (240 geometric features and 121 behavioral features), as summarized in Table 1.

Specifically, geometric features were extracted following the method described in [14]. These features include distance and angle between landmarks, mainly around eyebrows, eyes, and mouth, resulting in a total of 240 features as described in Table 1 and Figure 2.

In addition, we extracted 121 behavioral features from video data. In general, smiling is considered to be an expression of positive emotions, but it also has many other meanings [15], [64]. In fact, people may smile more often in negative or stressful situations [65], [66], [67]. Regarding this situation, to distinguish genuine from voluntary smiles, the 19th-century French neuroanatomist Duchenne de Boulogne proposed the concept of the Duchenne smile, which refers to natural laughter [16]. This concept allows us to distinguish between natural and non-natural smiles. The difference between Duchenne (natural) and non-Duchenne (non-natural) smiles has been validated in multiple fine-grained analyses [16], [68]. Zaalberg et al. [13] illustrate the difference between Duchenne and non-Duchenne smiles by

comparing facial expression responses in socially managed and unmanaged situations. Based on the observed differences in facial expressions, we extracted behavioral features that reflect the distinguishing characteristics of Duchenne and non-Duchenne smiles: the distance between the eyebrows, the distance between the eyes and eyebrows, the eye height, the jaw length, the mouth width and height, and the distance between the mouth and eyes.

To comprehensively capture emotional suppression beyond smile-related features, we additionally integrated behavioral features proposed in prior studies [2], [17]. These include body movements, facial movements, facial touching, changes in the eye position, and eye blinking. Since the dataset includes only the upper half of the subject's body, body movements were represented by changes in the shoulder position and angle. Facial movements and facial touching features were represented by the tilt angle of the face and the distance between the hand and the face. Additionally, to capture the temporal dynamic patterns of these behavioral signals, we extracted temporal features by computing the rate of change of each behavioral feature.

2) DETAILED FEATURE EXTRACTION METHODS FROM VIDEO DATA

We utilized the original recordings in the Hide-and-seek dataset [7]. To extract geometric and behavioral features from the video data, we employed MediaPipe [63] to detect 543 landmarks (33 pose landmarks, 468 face landmarks, and 21 landmarks per hand). Geometric features were calculated by identifying the MediaPipe landmarks that are spatially closest to the landmark points defined in the FERA 2015 model [69] (see Figure 2). Since the two landmark detection models differ in both the number and positioning of their landmarks, we overlaid the visualizations of both landmark sets and manually mapped each FERA 2015 point to its nearest MediaPipe landmark. Using these mapped pairs, we extracted the distances and angles between landmarks as defined in [14]. While the original study used 2D coordinates from only frontal face images, we calculated distances and angles using 3D coordinates, thereby accommodating non-frontal face orientations.

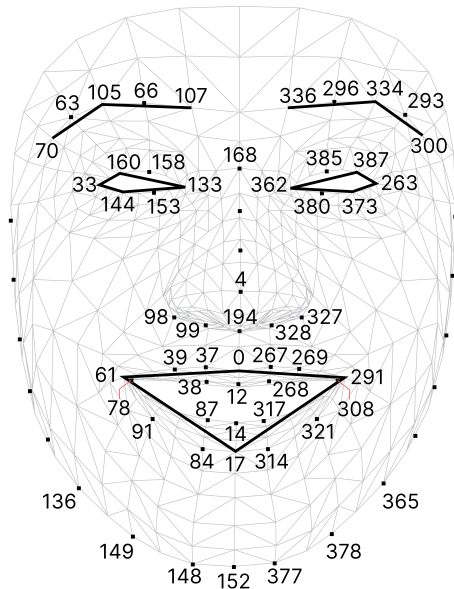


FIGURE 2. Visualization of Geometric Features. Following the method proposed by [14], we use facial landmark positions to extract geometric features such as the distance between the eyes, mouth, and eyebrows.

The details regarding the extraction of behavioral features are as follows:

- **Distance-based behavioral features.** Certain behavioral features were simply derived from distances between landmarks, as shown in Figure 2: (a) The distance between both eyebrows, (b) The distance between the centroid of both eyebrows and eyes, (c) The height of both eyes, (d) The length of jaw, (e) The width and the height of mouth, (f) The distance between the centroid of both eyes and mouth.
- **Body movement.** Body movement was measured as the variation in the center position of the shoulders and the

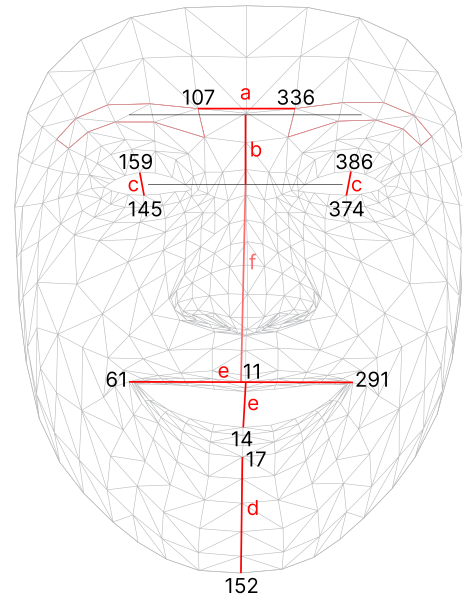


FIGURE 3. Visualization of Distance-based Behavioral Features. Each facial landmark position is used to extract the following distance-based behavioral features: (a) eyebrow distance, (b) eyebrow-to-eye center distance, (c) eye height, (d) jaw length, (e) mouth width and height, and (f) eye-to-mouth center distance.

change in the tilt angle of the shoulders with respect to a horizontal plane.

- **Facial movement.** Facial movement was measured as the tilt of the head with respect to a horizontal plane. For instance, if the subject holds their head upright, the measurement angle is approximately 90 degrees.
- **Facial touching.** Facial touching was measured as the distance between the midpoint of the face and the midpoint of the hand. If a single hand was detected, the distance to the face was measured. If both hands were detected, the shorter distance was used, as this is an indicator of how close the hands are to the face. If neither the hand nor the face was detected, the feature was imputed using the maximum distance observed from all the valid frames.
- **Eye movement.** Eye movement was measured as the degree of pupil shift by finding the relative position changes of the pupil within consecutive frames.
- **Eye blinking.** Eye blinking was detected when the average ratio of eye width to eye height across both eyes exceeded a threshold of 5. The value of 5 was chosen empirically through preliminary analysis.
- **Temporal features.** Temporal features were derived from the other behavioral signals and denoted them with the prefix *delta_*. Specifically, for each behavioral signal, we computed $a/|r|$, where a is the slope of a linear regression fitted over time within a 9-frame window and $|r|$ is the corresponding Pearson correlation coefficient [70]. a captures the rate of change, and $|r|$ reflects the consistency of trends within the window, yielding a reliability-weighted rate of change.

When the subject ducks or raises their head, the measured distance becomes distorted. For example, when a subject tilts their head downward and is captured by a frontal camera, the measured eye height may appear shorter than its actual value. To address this issue, we applied a correction by estimating head pose orientation and adjusting the corresponding distance values. To estimate the head orientation, we employed the solvePnP algorithm [71], which estimates the 3D pose of an object based on its 2D position. From each frame, we estimated the head orientation (pitch, yaw, and roll) of the worker from the 2D image in the video and categorized the head orientation as facing forward, downward, or upward. For non-frontal orientations (i.e., downward or upward), a pitch-angle-based correction was applied using the cosine of the pitch angle, as illustrated in Figure 2.

The video data were segmented into 5-second windows with a slide size of 1-second. For each segment, we extracted statistical features of the video features, such as the mean, the standard deviation, the minimum, and the maximum values, then we normalized them by using a StandardScaler. Unlike other features, blinking was additionally calculated as the number of blinks within the window size. As a result, a total of 240 geometric features and 121 behavioral features were obtained. Across the entire dataset (all modalities), we obtained 21,976 segments for the Given-Emotional-Workload label and 22,149 segments for the Perceived-Emotional-Workload label.

B. MODALITY FEATURE SET ABLATION

To assess the practical utility of different modality combinations, we evaluated model performance in three modality configurations. The modality configurations were as follows: (1) All modalities: This configuration utilizes all available data from audio, physiological, and video modalities to evaluate the overall performance of emotional workload detection. It was designed to investigate how much the addition of the video modality improves performance compared to previous studies that relied solely on audio and physiological signals [7].

(2) Features from easy-to-use physiological sensors: This configuration uses only the three physiological signals EDA, ACC, and TEMP, which are relatively easy to collect. Physiological data has been widely used for emotion recognition, stress detection, and emotion regulation research. However, the nature of data collection requires the attachment of various sensors, which is highly intrusive to the user. Therefore, it has been proposed to utilize only the data that can be easily obtained (i.e., collected from sensors that are less intrusive to the user) [7]. This condition aims to evaluate the practicality of using only easy-to-use physiological data and compare it with a video-only condition.

(3) Features from video data: This configuration uses only video-based features extracted from the worker's facial and body movements. The goal is to evaluate how well non-invasive visual data alone supports emotional workload

detection and to demonstrate the effectiveness of the proposed video features. These configurations were applied to machine learning models and an MKL model.

C. MODEL DESCRIPTION

1) MACHINE LEARNING & DEEP LEARNING

We utilized 7 widely used machine learning models to compare performance for different combinations of features: Support Vector Machine (SVM) [72], Decision Tree (DT) [73], Random Forest (RF) [74], AdaBoost (AB) [75], XGBoost (XGB) [76], Linear Discriminant Analysis (LDA) [77], and k-Nearest Neighbor (kNN) [78].

The SVM model was implemented with parameters adopted from [79], where the regularization parameter C was set to 107 and the kernel coefficient gamma was set to 0.001. Additionally, the XGB classifier was used with default parameters (i.e., a learning rate of 0.1 and a max depth of 3) [80]. For the remaining models, we followed the method in [23]. Specifically, for tree-based models (DT, RF, AB), information gain was used as the splitting criterion, and the minimum number of samples required to split an internal node was set to 20. The number of base estimators was set to 100 for RF and AB. For kNN, the number of neighbors k was set to 9.

We also trained modern deep learning models for video emotion recognition by treating video frames as input, including EfficientNet-B0 [81] and a CLIP-based classifier [82]. We used raw frames to account for non-frontal views and head-down postures that naturally occur during the task. Moreover, the background was fixed and controlled across sessions, minimizing potential background-related confounds. Both models were trained using the Adam optimizer and cross-entropy loss with a learning rate of 0.001 and a batch size of 32, for 12 and 15 epochs, respectively [42], [45]. For the CLIP-based model, we froze the embedding backbone and trained a classification head consisting of a 512-unit fully connected layer followed by a final classification layer [45]. Additionally, we trained an MLP on features extracted from ResNet50 [83], a commonly used feature extractor for video emotion recognition [38], [39], [40], [41]. The model architecture consisted of a single 32-unit hidden layer with ReLU activation followed by a sigmoid output layer, and it was trained using the SGD optimizer and cross-entropy loss [84] with a learning rate of 0.005 and 100 epochs [41].

2) MULTIPLE KERNEL LEARNING

To improve the performance of SVM-based emotion recognition models, we developed a framework using MKL (Multiple Kernel Learning), which assigns different feature types to separate kernels and combines them effectively. For example, in our case, features extracted from different modalities, such as audio and physiological signals, can be assigned to their respective kernel. Our MKL framework consists of three steps (see Figure 5). First, we assign a

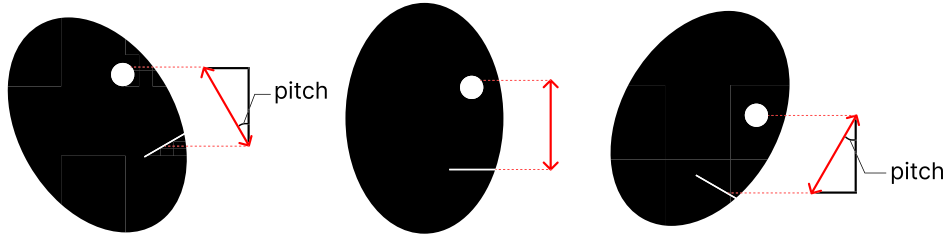


FIGURE 4. Correction of Facial Distance Distortion using Head Pose. When a subject tilts their head downward or upward, certain facial distances (e.g., eye-to-mouth center distance) appear distorted in the camera projection. In this example, the red arrow indicates the true distance between facial landmarks. To recover this value, we adjusted the observed (camera-projected) distance by multiplying the cosine of the estimated head pitch angle.

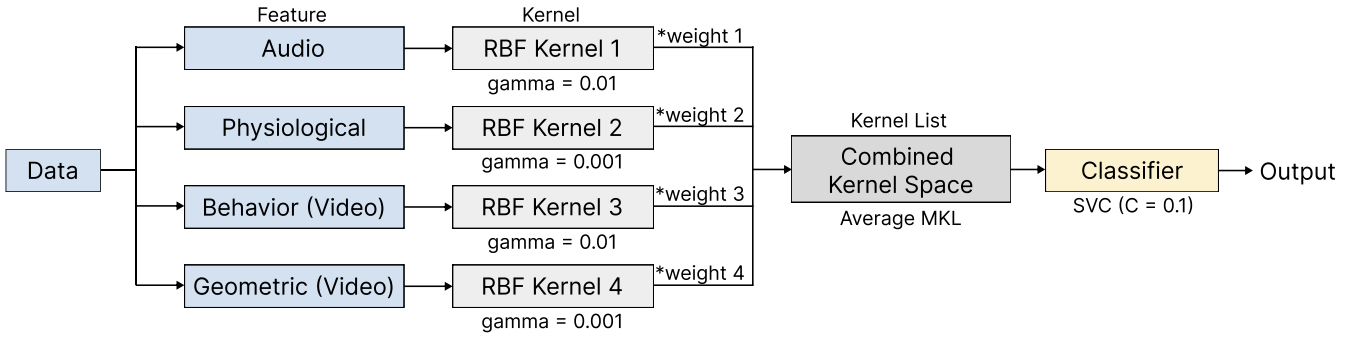


FIGURE 5. Multimodal Multiple Kernel Learning framework. Features from audio, physiological signals, and video are each mapped to separate RBF kernels with distinct hyperparameters. These kernels are combined using Average MKL, with learned weights. Finally, a Support Vector Classifier (SVC) is applied for final classification. In the case of the model using features from video data, features are divided into geometric, facial, and bodily behavioral features, each assigned to a separate kernel.

Gaussian kernel to each feature type. Second, we trained the weights for each kernel and combined them through a weighted linear combination. Third, we train an SVM classifier using the combined kernel.

Specifically, we assign a Gaussian kernel (RBF kernel) to each feature type and determine an appropriate gamma value for each. To compute the similarity between two samples, we use the following Gaussian kernel function, which assigns higher similarity to samples that are closer in the feature space based on their Euclidean distance.

$$\exp\left(-\gamma \|x - z\|_2^2\right) \quad (1)$$

where x and z are two samples (data points).

In Equation 1, γ is a scaling parameter that controls the similarity between samples. As the γ value increases, the model considers only nearby samples to be similar, which can lead to overfitting. Conversely, as the γ value decreases, the model treats even distant samples as similar, making it less sensitive but potentially causing underfitting. To find the optimal value of the hyperparameter γ , we performed a grid search over the set $\{0.0001, 0.001, 0.01, 0.1\}$.

In the next step, to combine each kernel with optimal weights, we quantify how well each kernel agrees with the target labels using the kernel alignment metric [18]. Specifically, for each kernel matrix K_m , we compute its

alignment with an *ideal kernel matrix* K_i ($K_i = yy^T$, where y is the binary label vector), using the inner product between K_m and K_i [85]. We denote this value as $\text{alignment}(K_m, K_i)$. We then normalize the alignment scores across all N kernels to obtain the kernel weight ratio r_m as follows:

$$r_m = \frac{\text{alignment}(K_m, K_i)}{\sum_{h=1}^N \text{alignment}(K_h, K_i)} \quad \forall m \in \{1, \dots, N\} \quad (2)$$

To avoid label leakage under the LOSO protocol, we compute the alignment-based weights r_m separately for each fold using only the training subjects (i.e., excluding the test subject and its labels). The resulting r_m are then applied to combine kernels when training and evaluating on that fold.

Finally, we assign each kernel a weight proportional to its normalized alignment score. To preserve the overall scale of the combined kernel (i.e., to match the unweighted case where all kernels have weight 1), we rescale the normalized weights by the number of kernels N . We then form the weighted kernels as $Nr_m K_m$ and combine these kernels using AverageMKL. We train the model using this combined kernel, with Support Vector Classification (SVC) as the base learner. To optimize the regularization parameter C of the classifier, we performed a grid search on the values $\{0.1, 1, 10, 100, 107\}$.

We aimed to improve the performance of both models using either all modalities or video-only data. First, in the model using all modalities, we applied separate

kernels for audio, physiological, behavioral, and geometric features, resulting in a total of four kernels. The grid search revealed that the optimal C value was 0.1 for the Perceived-Emotional-Workload label, and 107 for the Given-Emotional-Workload label. The gamma hyperparameters were individually optimized for each kernel. As a result, for the Perceived-Emotional-Workload label, the gamma value was set to 0.01 for audio and behavioral kernels, and 0.001 for physiological and geometric kernels. In the case of the Given-Emotional-Workload label, we found that a gamma of 0.01 performed best for audio features, and 0.0001 for behavioral, geometric, and physiological features.

In the model using features from video data, we first divided them based on the feature type (i.e., geometric and behavioral features), and then categorized them according to the location (i.e., face and body) from which they are extracted. As a result, we grouped the features into three categories, with a separate kernel applied to each: geometric features; facial behavioral features (i.e., distance-based behavioral features, facial movement, eye movement, and eye blinking) with their corresponding temporal features; and bodily behavioral features (i.e., body movement and facial movement) with their corresponding temporal features. The optimal C value of this model was 10 for the Perceived-Emotional-Workload label and 107 for the Given-Emotional-Workload label. These values are the same as those of the model using all feature types. For the Perceived-Emotional-Workload label, the optimal gamma value was 0.01 for bodily behavioral features, and 0.0001 for facial behavioral features and geometric features. For the Given-Emotional-Workload label, the optimal gamma value was 0.0001 for all feature sets.

D. EVALUATION

We evaluated model performance using the Leave-One-Subject-Out (LOSO) cross-validation method. Since the LOSO cross-validation method excludes one participant per fold, it allows for a rigorous evaluation of the model's generalization ability to unseen users, thereby reducing the risk of performance overestimation. For the binary classification tasks of Given-Emotional-Workload and Perceived-Emotional-Workload labels, we employed accuracy, balanced accuracy, AUC, and F1-score as evaluation metrics. To ensure robustness of machine learning and MKL models, we train the model five times and report the mean and standard deviation of the metrics.

V. RESULTS

We conducted binary classification tasks for Given-Emotional-Workload and Perceived-Emotional-Workload labels using three different feature configurations (all modalities, features from easy-to-use physiological sensors, and features from video data). This resulted in a total of six classification settings. The performance of models for each configuration is presented in Table 2, 3, 4, 5, and 6. For subsequent analysis, we focused primarily on the SVM model. It was selected as the baseline model due

to its scalability and suitability for multimodal integration, particularly with kernel-based approaches such as multi-kernel learning.

We also analyzed the feature importance in the model with all modalities and features from video data to identify the most informative video features for detecting emotional workload. To extract the feature importance of the SVM classifier, we used the *permutation_importance* function provided by scikit-learn [86]. Permutation feature importance is a model inspection technique that calculates the contribution of each feature based on the model's statistical performance change when a specific feature value is randomly permuted. If the performance of the model decreases when a specific feature is randomly permuted, it means that the feature is given a higher feature importance value. Numerically, if a feature has a feature importance value of 0.02, the model's performance decreased by 2% when the feature was randomized. The feature importance shown in Figure 6 and Figure 7 is the average of the importance values assigned to each feature over the five iterations. The y-axis represents each feature, and the x-axis represents the importance value of the feature. The y-axis is sorted in order of increasing feature importance.

Figure 6 shows the top 20 features with the highest correlations, based on all features extracted from audio, physiological, and video data. The top 20 features were mostly derived from customer voice data, followed by worker video data. Figure 7 shows the top 20 highly correlated features from the model using video data. The top 20 features were primarily geometric features obtained from worker video data. The interpretation of feature importance is discussed in detail in the Discussion section.

A. EFFECTIVENESS OF VIDEO FEATURES

To assess the contribution of features extracted from video data, we compared the emotional workload detection performance of SVM-based models trained with all modalities (including video features) and those trained without video features. As shown in Table 5, for the Perceived-Emotional-Workload label, the model incorporating video features showed higher accuracy and balanced accuracy (accuracy = 0.66, $SD = 0.09$; balanced accuracy = 0.67, $SD = 0.10$) than the model without video features (accuracy = 0.65, $SD = 0.11$; balanced accuracy = 0.65, $SD = 0.11$). In contrast, for the Given-Emotional-Workload label, the model with video features achieved the same accuracy (0.87, $SD = 0.13$) and slightly lower balanced accuracy (0.85, $SD = 0.14$) compared to the model without video features (balanced accuracy = 0.86, $SD = 0.14$). This may be because audio-related features were particularly informative for the Given-Emotional-Workload label. They ranked highly in the feature importance list, and in the data collection scenario, the audio features (particularly the customer's voice) served as situational cues for the Given-Emotional-Workload label, functioning as a criterion for

TABLE 2. Evaluation of the Classifiers on Multimodal Data: Comparison of the machine learning models using all multimodal data for detecting Given- and Perceived-Emotional-Workload. Reported values are mean (SD) over five independent training runs. Bal Acc denotes balanced accuracy.

Model	Given-Emotional-Workload				Perceived-Emotional-Workload			
	Accuracy	Bal Acc	F1-score	AUC score	Accuracy	Bal Acc	F1-score	AUC score
SVM	0.87 (0.13)	0.85 (0.14)	0.90 (0.10)	0.93 (0.11)	0.66 (0.09)	0.67 (0.10)	0.66 (0.11)	0.73 (0.13)
Decision Tree	0.80 (0.13)	0.78 (0.13)	0.84 (0.10)	0.79 (0.13)	0.60 (0.09)	0.60 (0.09)	0.61 (0.11)	0.62 (0.10)
Random Forest	0.86 (0.14)	0.84 (0.16)	0.90 (0.10)	0.92 (0.12)	0.70 (0.11)	0.71 (0.12)	0.70 (0.13)	0.78 (0.14)
AdaBoost	0.86 (0.17)	0.82 (0.20)	0.90 (0.12)	0.91 (0.15)	0.69 (0.10)	0.70 (0.11)	0.70 (0.11)	0.77 (0.14)
XGBoost	0.85 (0.16)	0.83 (0.18)	0.89 (0.13)	0.91 (0.15)	0.70 (0.11)	0.70 (0.11)	0.69 (0.13)	0.78 (0.15)
LDA	0.85 (0.11)	0.84 (0.12)	0.88 (0.09)	0.91 (0.11)	0.67 (0.13)	0.67 (0.13)	0.67 (0.14)	0.74 (0.16)
KNN	0.76 (0.07)	0.72 (0.08)	0.82 (0.05)	0.82 (0.08)	0.63 (0.08)	0.63 (0.08)	0.65 (0.09)	0.69 (0.11)
Majority Voting	0.66 (0.01)	0.50 (0.00)	0.80 (0.01)	0.50 (0.00)	0.58 (0.06)	0.50 (0.00)	0.43 (0.37)	0.50 (0.00)

TABLE 3. Evaluation of the Classifiers on Easy-to-use Physiological Signals: Comparison of the machine learning models using EDA, ACC, and TEMP signals for detecting Given- and Perceived-Emotional-Workload. Reported values are mean (SD) over five independent training runs. Bal Acc denotes balanced accuracy.

Model	Given-Emotional-Workload				Perceived-Emotional-Workload			
	Accuracy	Bal Acc	F1-score	AUC score	Accuracy	Bal Acc	F1-score	AUC score
SVM	0.65 (0.14)	0.55 (0.14)	0.76 (0.13)	0.62 (0.25)	0.54 (0.13)	0.55 (0.12)	0.57 (0.14)	0.60 (0.18)
Decision Tree	0.62 (0.12)	0.57 (0.14)	0.71 (0.11)	0.58 (0.15)	0.52 (0.11)	0.52 (0.10)	0.49 (0.16)	0.52 (0.11)
Random Forest	0.64 (0.15)	0.57 (0.16)	0.74 (0.12)	0.63 (0.23)	0.55 (0.13)	0.56 (0.13)	0.52 (0.19)	0.59 (0.19)
AdaBoost	0.64 (0.14)	0.54 (0.14)	0.75 (0.10)	0.60 (0.25)	0.56 (0.13)	0.57 (0.13)	0.55 (0.16)	0.60 (0.19)
XGBoost	0.65 (0.15)	0.57 (0.16)	0.75 (0.13)	0.63 (0.25)	0.50 (0.12)	0.52 (0.11)	0.49 (0.19)	0.56 (0.19)
LDA	0.67 (0.15)	0.58 (0.16)	0.76 (0.15)	0.66 (0.28)	0.58 (0.15)	0.58 (0.14)	0.59 (0.16)	0.63 (0.20)
KNN	0.59 (0.14)	0.51 (0.14)	0.70 (0.11)	0.52 (0.21)	0.51 (0.09)	0.52 (0.08)	0.51 (0.11)	0.51 (0.12)
Majority Voting	0.66 (0.01)	0.50 (0.00)	0.80 (0.01)	0.50 (0.00)	0.58 (0.06)	0.50 (0.00)	0.43 (0.37)	0.50 (0.00)

TABLE 4. Evaluation of the Classifiers on Features from Video data: Comparison of the machine learning models using geometric and behavioral features and deep learning models using video frames, for detecting Given- and Perceived-Emotional-Workload. Values are mean (SD): machine learning over five runs; deep learning over LOSO folds (final epoch). Bal Acc denotes balanced accuracy.

Model	Given-Emotional-Workload				Perceived-Emotional-Workload			
	Accuracy	Bal Acc	F1-score	AUC score	Accuracy	Bal Acc	F1-score	AUC score
SVM	0.63 (0.09)	0.58 (0.09)	0.73 (0.08)	0.63 (0.14)	0.59 (0.09)	0.59 (0.09)	0.59 (0.12)	0.62 (0.13)
Decision Tree	0.59 (0.05)	0.54 (0.06)	0.68 (0.05)	0.55 (0.07)	0.54 (0.05)	0.54 (0.06)	0.53 (0.08)	0.55 (0.06)
Random Forest	0.67 (0.05)	0.55 (0.06)	0.79 (0.04)	0.66 (0.14)	0.62 (0.09)	0.62 (0.09)	0.63 (0.10)	0.67 (0.12)
AdaBoost	0.66 (0.03)	0.52 (0.03)	0.79 (0.02)	0.64 (0.12)	0.61 (0.09)	0.61 (0.09)	0.63 (0.10)	0.66 (0.12)
XGBoost	0.67 (0.07)	0.57 (0.08)	0.78 (0.05)	0.64 (0.15)	0.61 (0.10)	0.61 (0.10)	0.62 (0.11)	0.65 (0.13)
LDA	0.65 (0.09)	0.59 (0.11)	0.75 (0.07)	0.63 (0.14)	0.61 (0.10)	0.61 (0.10)	0.63 (0.10)	0.65 (0.13)
KNN	0.62 (0.06)	0.54 (0.07)	0.73 (0.05)	0.56 (0.10)	0.57 (0.07)	0.57 (0.07)	0.60 (0.09)	0.60 (0.10)
EfficientNet	0.57 (0.17)	0.50 (0.14)	0.66 (0.22)	0.51 (0.23)	0.52 (0.08)	0.52 (0.07)	0.41 (0.20)	0.53 (0.11)
CLIP	0.56 (0.11)	0.53 (0.06)	0.61 (0.23)	0.57 (0.11)	0.49 (0.08)	0.49 (0.06)	0.33 (0.20)	0.49 (0.10)
MLP	0.52 (0.14)	0.47 (0.09)	0.59 (0.25)	0.45 (0.17)	0.48 (0.08)	0.47 (0.06)	0.33 (0.25)	0.43 (0.10)
Majority Voting	0.66 (0.01)	0.50 (0.00)	0.80 (0.01)	0.50 (0.00)	0.58 (0.06)	0.50 (0.00)	0.43 (0.37)	0.50 (0.00)

TABLE 5. Effectiveness of MKL for Multimodal Feature Fusion: Evaluation of SVM and MKL-based models trained on all available modalities, and comparison of performance with and without video features for detecting Given- and Perceived-Emotional-Workload. Reported values are mean (SD) over five independent training runs. Bal Acc denotes balanced accuracy.

Model	Data Combination	Given-Emotional-Workload				Perceived-Emotional-Workload			
		Accuracy	Bal Acc	F1-score	AUC score	Accuracy	Bal Acc	F1-score	AUC score
SVM	Physiological + Audio + Video	0.87 (0.13)	0.85 (0.14)	0.90 (0.10)	0.93 (0.11)	0.66 (0.09)	0.67 (0.10)	0.66 (0.11)	0.73 (0.13)
	Physiological + Audio	0.87 (0.13)	0.86 (0.14)	0.90 (0.11)	0.92 (0.12)	0.65 (0.11)	0.65 (0.11)	0.64 (0.13)	0.71 (0.15)
	Audio	0.84 (0.17)	0.82 (0.19)	0.88 (0.13)	0.90 (0.15)	0.66 (0.11)	0.67 (0.11)	0.66 (0.11)	0.72 (0.14)
MKL	Physiological + Audio + B.video + G.video	0.88 (0.11)	0.87 (0.12)	0.91 (0.09)	0.94 (0.09)	0.71 (0.11)	0.72 (0.12)	0.72 (0.12)	0.79 (0.14)
Majority Voting		0.66 (0.01)	0.50 (0.00)	0.80 (0.01)	0.50 (0.00)	0.58 (0.06)	0.50 (0.00)	0.43 (0.37)	0.50 (0.00)

distinguishing between the stimulated and unstimulated call sections. Accordingly, adding video or other modalities may not have led to substantial improvements for this label.

In addition, to verify the external validity and scalability of video features (i.e., behavioral features and geometric features), we compared the performance of the model using

features from video data with that of the model using features from easy-to-use physiological sensors (i.e., easily available physiological signals). We also compared it with the model using all physiological features, which require sensor attachment during data collection, to assess whether video data could even outperform all physiological input.

TABLE 6. Effectiveness of MKL for Video Feature Fusion: Evaluation of SVM and MKL-based models trained on features from video data (i.e., behavioral and geometric), with performance compared against models using all and easy-to-use physiological features for detecting Given- and Perceived-Emotional-Workload. Reported values are mean (SD) over five independent training runs. Bal Acc denotes balanced accuracy.

Model	Data Combination	Given-Emotional-Workload				Perceived-Emotional-Workload			
		Accuracy	Bal Acc	F1-score	AUC score	Accuracy	Bal Acc	F1-score	AUC score
SVM	Physiological	0.69 (0.12)	0.63 (0.13)	0.78 (0.10)	0.69 (0.18)	0.59 (0.11)	0.59 (0.10)	0.59 (0.14)	0.64 (0.14)
	Physiological(EDA, ACC, TEMP)	0.65 (0.14)	0.55 (0.14)	0.76 (0.13)	0.62 (0.25)	0.54 (0.13)	0.55 (0.12)	0.57 (0.14)	0.60 (0.18)
	Video	0.63 (0.09)	0.58 (0.09)	0.73 (0.08)	0.63 (0.14)	0.59 (0.09)	0.59 (0.09)	0.59 (0.12)	0.62 (0.13)
MKL	facial B.video + bodily B.video + G.video	0.67 (0.09)	0.59 (0.10)	0.77 (0.07)	0.65 (0.15)	0.63 (0.09)	0.63 (0.10)	0.65 (0.09)	0.67 (0.13)
	Majority Voting	0.66 (0.01)	0.50 (0.00)	0.80 (0.01)	0.50 (0.00)	0.58 (0.06)	0.50 (0.00)	0.43 (0.37)	0.50 (0.00)

As a result, for the Given-Emotional-Workload label, the model using video features achieved lower accuracy and balanced accuracy (accuracy = 0.63, $SD = 0.09$; balanced accuracy = 0.58, $SD = 0.09$) than the model using all physiological features (accuracy = 0.69, $SD = 0.12$; balanced accuracy = 0.63, $SD = 0.13$). Compared with the model using features from easy-to-use physiological sensors (accuracy = 0.65, $SD = 0.14$; balanced accuracy = 0.55, $SD = 0.14$), the model using video features showed lower accuracy but slightly higher balanced accuracy.

Interestingly, for the Perceived-Emotional-Workload label, the model using video features achieved the highest performance, with both accuracy and balanced accuracy of 0.59, tied with the model using all physiological features. As shown in Table 6, these values were 0.05 higher in accuracy and 0.04 higher in balanced accuracy than those of the model using features from easy-to-use physiological sensors. In other words, video features alone were able to achieve the same performance as widely used but highly invasive physiological features. Furthermore, video features outperformed the method based on readily available physiological data (EDA, ACC, TEMP), which was proposed to compensate for the shortcomings of physiological features. These results showed that emotional workload can be estimated with improved usability and accuracy using only video features without invasive wearable sensor data.

We also tested video-only deep learning baselines, but they did not perform competitively. Under the LOSO setting, the deep-learning models may have learned subject-specific visual patterns from the training subjects (e.g., facial shape or hairstyle), which may not generalize well to the test subject. Although collecting data from human subjects involves costs and practical constraints, securing larger and more diverse datasets is necessary to mitigate inter-subject variability and improve generalization.

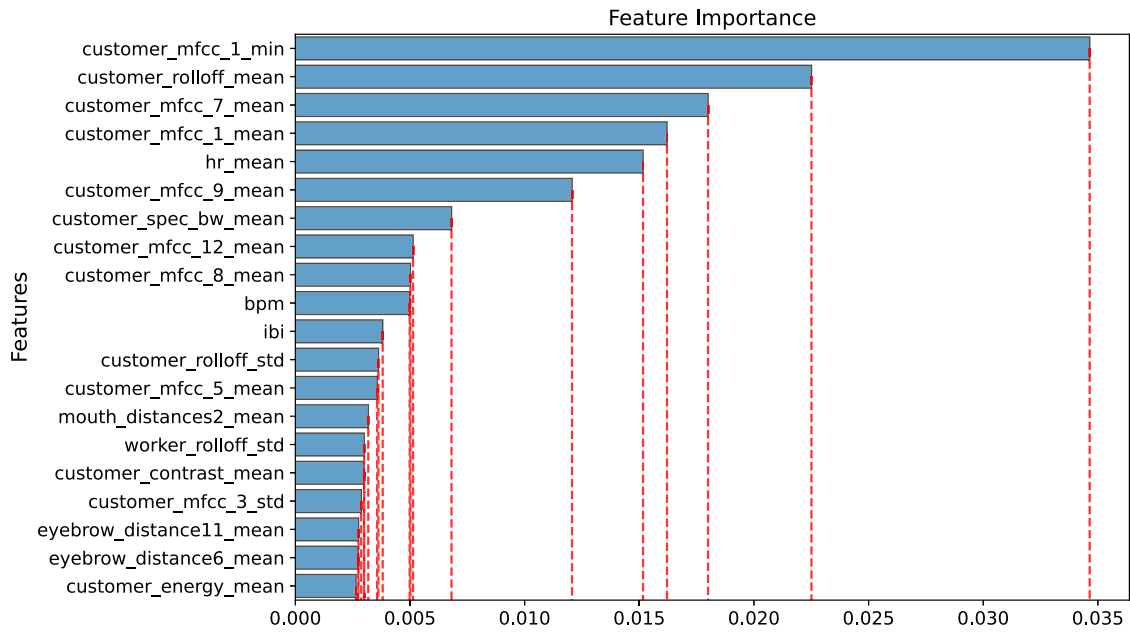
B. EFFECTIVENESS OF MULTIMODAL FUSION USING MULTIPLE KERNEL LEARNING

We applied MKL to improve the performance of two models: one using all modalities and the other using features from video data. Additionally, to examine the effectiveness of MKL in detecting emotional workload, we compared its performance with that of several widely used machine learning and deep learning models. For the

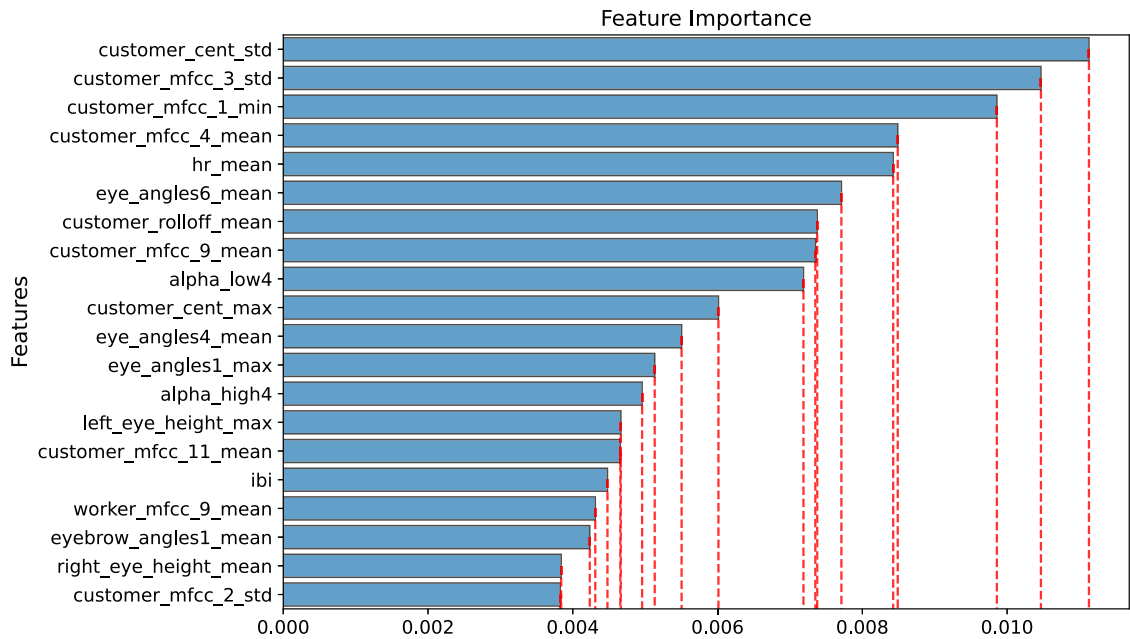
Given-Emotional-Workload label, the optimized MKL-based model using all feature types (i.e., behavioral, geometric, audio, and physiological) showed an accuracy of 0.88 ($SD = 0.11$), a balanced accuracy of 0.87 ($SD = 0.12$), an F1-score of 0.91 ($SD = 0.09$), and an AUC of 0.94 ($SD = 0.09$). Compared to the single-kernel SVM, MKL improved accuracy and balanced accuracy by 0.01 and 0.02, respectively (both $p < .01$, paired Wilcoxon signed-rank tests on per-subject LOSO results). For the Perceived-Emotional-Workload label, the MKL-based model using all modalities also outperformed the single-kernel SVM model, improving accuracy and balanced accuracy by 0.05 (accuracy = 0.71, $SD = 0.11$; balanced accuracy = 0.72, $SD = 0.12$) and improving both F1-score and AUC by 0.06 (F1 = 0.72, $SD = 0.12$; AUC = 0.79, $SD = 0.14$). Moreover, MKL achieved slightly higher accuracy and balanced accuracy (by 0.01) than the best-performing machine learning model (see Table 2).

In the MKL-based model using features from video data, substantial performance improvements for both Given- and Perceived-Emotional-Workload labels were observed compared to the single-kernel SVM model, as shown in Table 6. Specifically, for the Given-Emotional-Workload label, the MKL-based model using features from video data achieved an accuracy of 0.67, a balanced accuracy of 0.59, and an F1-score of 0.77, showing improvements of 0.04, 0.01, and 0.04, respectively. For the Perceived-Emotional-Workload label, the MKL-based model using features from video data achieved an accuracy of 0.63, a balanced accuracy of 0.63, and an F1-score of 0.64, with improvements of 0.04, 0.04, and 0.06, respectively.

Furthermore, the MKL-based model using video data outperformed the SVM model using less intrusive physiological data (EDA, ACC, TEMP), improving accuracy by 0.02 and 0.09 and balanced accuracy by 0.04 and 0.08 for Given- and Perceived-Emotional-Workload labels, respectively. Notably, for the Perceived-Emotional-Workload label, it also achieved 0.04 higher accuracy and balanced accuracy than the SVM model using all physiological features. In addition, its accuracy was slightly higher by 0.01 than the highest accuracy achieved by other machine learning models using either video or easy-to-use physiological features (EDA, ACC, TEMP) (see Table 3 and Table 4). Beyond high predictive performance, MKL offers several structural advantages: it allows assigning separate kernels



(a) Feature importance of model using all multimodalities for the Given-Emotional-Workload label



(b) Feature importance of model using all multimodalities for the Perceived-Emotional-Workload label

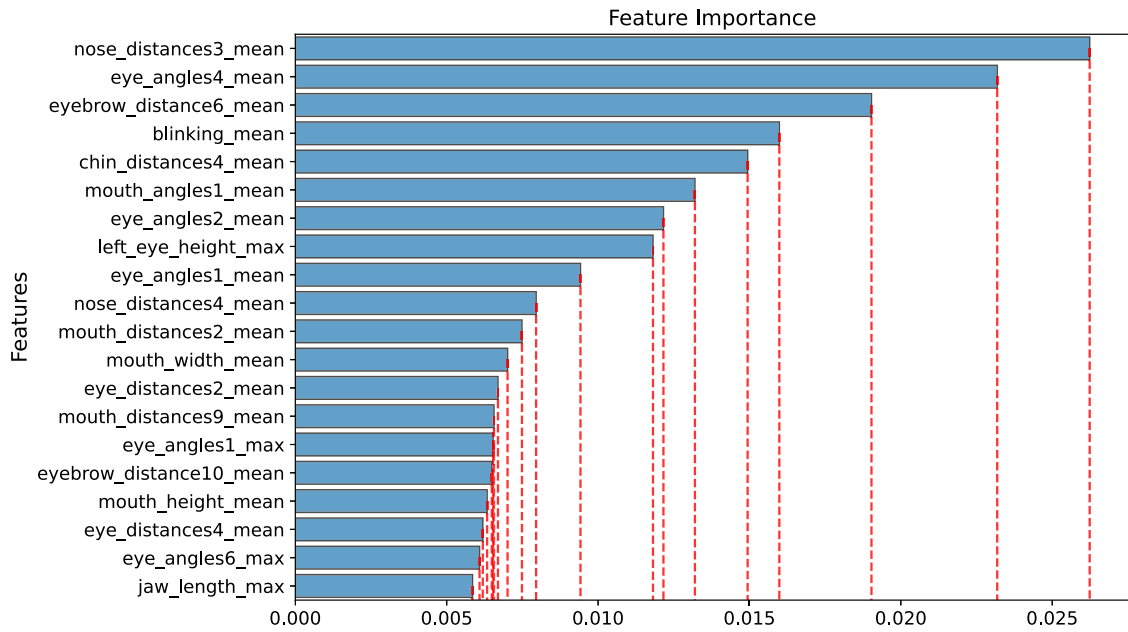
FIGURE 6. The Top 20 Multimodal Features Based on Permutation Feature Importance: Evaluation of the most influential features among all multimodal inputs using permutation importance with an SVM classifier. Importance scores are shown separately for (a) Given- and (b) Perceived-Emotional-Workload labels.

to each feature group, enabling more precise modeling of feature-specific characteristics. This modular architecture enhances flexibility by allowing integration of heterogeneous data types, supports customization by enabling independent adjustment of each feature’s influence. It also improves scalability by enabling new feature groups to be incorporated without full retraining, as kernels trained on existing feature groups can be reused. It is particularly well-suited for tasks

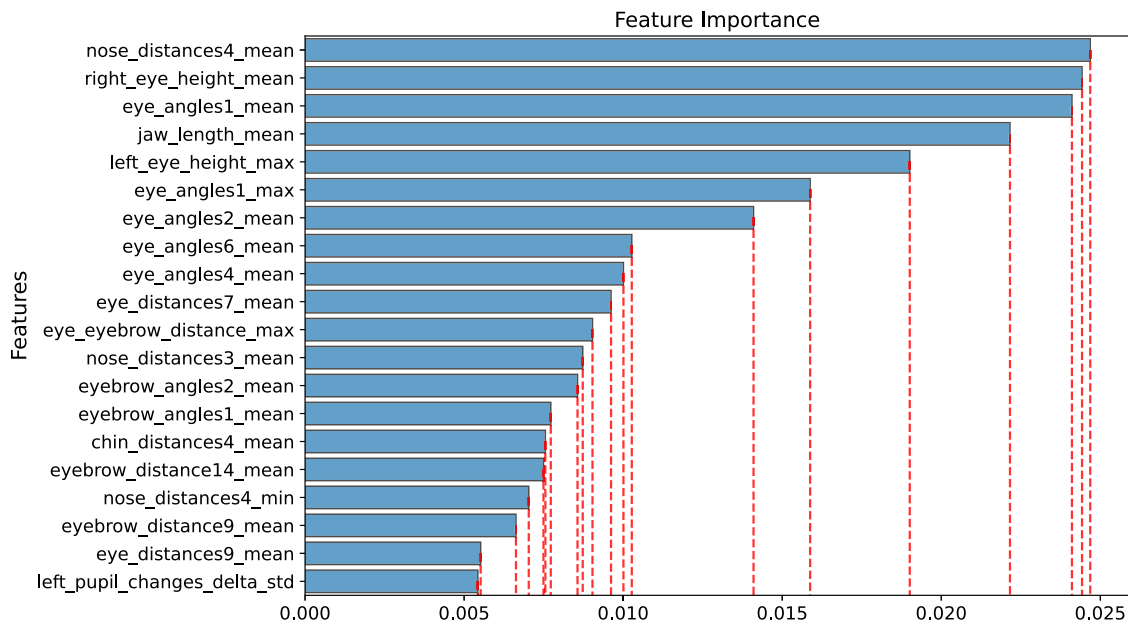
where separate representations may be required for each input feature type.

VI. DISCUSSION

In this section, we discuss the factors contributing to the model’s performance in emotional workload detection and examine its external validity and scalability.



(a) Feature importance of model using features from video data for the Given-Emotional-Workload label



(b) Feature importance of model using features from video data for the Perceived-Emotional-Workload label

FIGURE 7. The Top 20 Video Features Based on Permutation Feature Importance: Evaluation of the most influential features among video inputs (geometric and behavioral features) using permutation importance with an SVM classifier. Importance scores are shown separately for (a) Given- and (b) Perceived-Emotional-Workload labels.

A. WHICH VIDEO FEATURES WERE EFFECTIVE FOR EMOTIONAL WORKLOAD DETECTION?

We demonstrate the effectiveness of the video-based features. In the following analysis, we specifically examine which features contributed most to the improvement in model performance. In the model utilizing all multimodal data, geometric video features are at the top of the feature

importance list for both the Given-Emotional-Workload and the Perceived-Emotional-Workload labels. Notably, geometric features related to eyebrows appeared among the top-ranked features for both labels. This suggests that, as with [13], there were differences in the shape of the eyebrow when suppression was given or not. Interestingly, we observed user behaviors such as frowning, which led

to changes in eyebrow angle during the experiment. For the Perceived-Emotional-Workload label, geometric features related to the eye were also particularly prevalent. This aligns with prior findings [14], [87], [88], [89], suggesting that eye shape or movement differ depending on whether emotional suppression was presented. Furthermore, we observed that users looked at the floor or closed their eyes when stimulated during the experiment.

Additionally, for the Perceived-Emotional-Workload label, `left/right_eye_height_mean` ranked among the top features. This may be because the Perceived-Emotional-Workload label is self-reported and thus may reflect individual affective experiences more directly, making subtle facial and behavioral cues relatively more informative than for the condition-defined Given-Emotional-Workload label. These behavioral features capture cues associated with non-Duchenne smiles. Specifically, it reflects Lid Tightener, Action Unit 7 (AU 7) of the Facial Action Coding System (FACS) [90]. Lid Tightener is characterized by changing the height of the eyes, such as narrowing the eyes. The high importance of this feature suggests that cues known to differ between non-Duchenne and Duchenne smiles were informative for detecting emotional workload.

In the model utilizing only video data, similar to all multimodal data, geometric video features dominantly rank at the top of the feature importance list for both labels. This is likely because geometric features are easier to generalize than behavioral features, which can highly reflect individual differences such as personal habits. Behavioral features are likely to reflect individual behavioral differences, such as personal habits, making them more variable and harder to generalize across users. In contrast, geometric features, though influenced by minor individual differences in facial structure, such as the position of the eyes, nose, and mouth, tend to be more consistent across users.

Nonetheless, behavioral features also appeared among the top 10, including `blinking_mean` and `left/right_eye_height_max` for the Given-Emotional-Workload label, and `right_eye_height_mean`, `jaw_length_mean`, and `left_eye_height_max` for the Perceived-Emotional-Workload label.

Similar to `left_eye_height`, `jaw_length` was derived from AUs associated with non-Duchenne smiles, as opposed to natural Duchenne smiles. Specifically, the jaw length was calculated as the distance between the lowest point of the lips and the lowest point of the face by considering AU 15 and AU 17. The AU 15, Lip Corner Depressor, is the behavior of pulling the corners of the mouth down. Also, the AU 17, Chin Raiser, is the behavior of lifting the jaw up to make the lips protrude. These AUs are associated with changes in jaw length. The prominence of these features also suggests that cues known to differ between non-Duchenne and Duchenne smiles were informative for detecting emotional workload. Additionally, the high rank of `blinking_mean` indicates that when the user is suppressed, changes in blinking frequency are revealed

as in stress [17], [91], [92], [93]. This result is also consistent with [27], which reported that blinking was helpful for the detection of hidden emotion.

B. WHY DOES MKL PERFORM BETTER FOR SUBJECTIVE EMOTIONAL WORKLOAD?

We demonstrated that MKL enhances the performance of the emotional workload detection model. While the MKL-based model generally achieved higher accuracy on the Given-Emotional-Workload label, it showed a greater performance improvement on the Perceived-Emotional-Workload label. This suggests that MKL effectively learned the latent pattern in the Perceived-Emotional-Workload label. The Given-Emotional-Workload label was objectively assigned based on predefined experimental conditions (i.e., neutral vs. shouting or swearing); there is an explicit correlation between the customer's audio data (i.e., the situational cue) and the predefined label. In contrast, the Perceived-Emotional-Workload label was subjective, based on each participant's self-reported emotional state. Self-reports can be influenced by various factors, such as the ability to express emotions, personality, and emotional granularity [94], introducing latent patterns that are difficult to observe explicitly. Notably, subjective labels are more likely to include complex factors affecting emotional workload, resulting in a complex and non-linear relationship between data and labels. Models trained on self-reported labels generally exhibit lower performance due to their complex and non-linear relationship. Therefore, it is noteworthy that the MKL-based model led to improved detection performance on self-reported labels.

MKL assigns different kernels suited to the characteristics of each feature type and combines them with appropriate weights, enabling it to more effectively capture complex patterns, such as the relationships between subjective labels and data. In this process, it is important to optimize the hyperparameters for each kernel individually, as this allows better preservation of the characteristics of individual features. The process of combining these different kernels is also critical for optimal performance. In our experiments, we found that a weighted linear combination performed better than both non-linear and unweighted linear combinations. Thus, we assigned weights to each kernel based on their relationship, enabling a more balanced integration of various feature types. In this way, MKL provides greater flexibility in capturing complex feature patterns, particularly those that are difficult to observe explicitly.

C. EXTERNAL VALIDITY AND SCALABILITY

We also examined the external validity and scalability of our proposed framework, which detects emotional workload. To address this, we compared the performance of three SVM-based models: the model using features from video data, the model using all physiological data, and the model using features from easy-to-use physiological sensors.

As shown in Table 6, the accuracy of detecting emotional workload of each model for the Perceived-Emotional-Workload label is higher in the following order: the model using features from video data, the model using all physiological data, and the model using features from easy-to-use physiological sensors. Notably, the model using non-invasive video data, which utilized only the camera to detect the user's state, showed improved accuracy compared to the model using less invasive physiological data (EDA, ACC, TEMP), and achieved accuracy levels comparable to the model using all physiological data. Furthermore, we improved the performance of the video-based model by incorporating MKL. Collecting physiological data typically requires users to attach multiple wearable devices during the experiment, which can be burdensome. In contrast, collecting video data does not require users to wear any additional wearable devices, offering a more convenient and less intrusive alternative [95], [96]. This approach requires only a single camera to capture the user's face and body. Therefore, the video-based sensing approach can provide a scalable and non-invasive solution for emotional workload detection in the real world due to its low setup requirements.

However, there are several points to consider regarding the video-based approach. The performance of the video-based approach can degrade in real-world conditions (e.g., variable lighting or occlusion) that differ from the Hide-and-seek data-collection setting, where lighting was relatively stable and faces were consistently visible. Moreover, users may exhibit unconscious inhibitory behavior when they are aware of being recorded, reducing the reliability of features such as facial expressions and eye blinks. With prolonged use, users may develop habits that suppress or distort emotional expressions, potentially reducing feature stability over time.

Nevertheless, a video-based setup is attractive for scalable, low-burden deployment because it requires only a single camera. It can also support a range of scalable, contactless applications (e.g., low-cost monitoring of remote workers' affective states or providing feedback on online learners' stress levels) with minimal setup and minimal interference with user activity. From a practical standpoint, when cameras are not acceptable due to privacy concerns or workplace policies, easy-to-use physiological sensors (EDA, ACC, TEMP) provide a feasible and relatively low-intrusive alternative.

D. ETHICAL CONSIDERATIONS

If the emotional workload experienced by workers persists, it can cause serious mental problems for workers. Therefore, we proposed a real-world applicable emotional workload detection model, protecting the well-being of workers. However, our model still has the potential to raise ethical concerns for users. Firstly, even if the emotional workload detection model can be used to enhance well-being with the user's consent, the perception of being monitored and having their data collected can cause mental distress. Also, the collection of data itself raises serious privacy concerns.

In particular, physiological data can be used to estimate a worker's health status, monitor worker productivity, and thereby directly influence the hiring process. Audio data and video data may also contain personal information, such as a home address and a worker's face image. However, instead of storing videos or photos that reveal a subject's appearance, we can extract features directly from video data by extracting landmarks, making it more difficult to infer the user's privacy. To further strengthen the privacy protection of data, data can be encrypted using privacy-preserving techniques such as homomorphic encryption.

Therefore, several ethical and legal guidelines must be followed. First, during data collection, it is essential to ensure that workers fully understand the purpose and method of using their personal data with their explicit informed consent, and immediately stop if they choose not to participate. Second, access to the data must be strictly restricted to authorized personnel and used solely to support employees' well-being (e.g., enabling early detection of burnout and timely support), not for surveillance, productivity tracking, or performance evaluation. Moreover, deploying emotion AI in workplace settings beyond research (e.g., for commercialization) can be ethically problematic and may be legally restricted in some countries, so relevant legal frameworks must be carefully considered.

VII. LIMITATIONS

A. DATA COLLECTION BIAS AND SMALL SAMPLE SIZE

Hide-and-seek dataset exhibited an imbalance in the gender ratio of worker participants, with a higher proportion of females. However, the proportion of female workers was overwhelmingly high in the actual call center environment. In addition, our study involved a relatively small sample size (31 participants) and was conducted within a single cultural context, which may limit generalizability and increase the risk of overfitting. In future work, building and leveraging larger and more diverse datasets will be an important avenue for research, and our findings should be validated using such datasets.

B. CONCERNS ABOUT EMOTIONAL REALISM IN THE IN-LAB SCENARIO

Since the experiment was based on a simulated scenario, participants' emotional responses may not have been as natural or intense as those elicited in real-world situations. Nevertheless, participants reported a high level of immersion during the experiments, suggesting that the scenario still elicited meaningful affective engagement.

C. LOSS OF VIDEO FEATURES

Due to the lighting condition, head posture, or face-covering behaviors during data collection, the participant's face was not detected in some video frames, resulting in the loss of video features. To address this, we applied interpolation by

averaging the values of the preceding and following frames where the facial landmark was not detected.

VIII. CONCLUSION

In this study, we proposed a non-invasive approach to emotional workload detection by utilizing video-based features and applying MKL for effective multimodal fusion. Specifically, we extracted geometric and behavioral features from video data and demonstrated that video features significantly contributed to model performance. Furthermore, the MKL approach enabled the model to capture the latent patterns between various features and labels, leading to significant accuracy improvements in emotional workload classification. Our findings highlight the potential of video-based, non-invasive methods as scalable and practical for the emotional workload model in real-world settings.

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